

Central sensitization, work-related stress, and musculoskeletal pain symptoms in desk-based workers with and without migraine: a cross-sectional study

Background Migraine is a common neurological disorder that causes disability worldwide and constitutes a significant individual and societal burden, particularly in working-age individuals, as it can substantially impact work performance, productivity, and daily living activities [1]. Central sensitization (CS) mechanisms play a role in the pathophysiology and chronicity of migraine, leading to increased pain sensitivity and potentially associated with musculoskeletal pain [2 , 3]. Furthermore, psychosocial factors and work stress have been identified as important factors associated with both headaches and musculoskeletal pain [1 , 3]. Literature reports that neck pain (NP) is common in individuals with migraine headaches; in fact, NP is twice as frequent in patients with chronic migraine (CM) headaches compared to those with episodic migraine (EM) [4 – 6]. While evidence for low back pain (LBP) is limited, it has been shown that individuals with chronic headaches report LBP more frequently [7]. In this context, the impact of work-related factors on migraine symptoms and accompanying musculoskeletal problems is becoming increasingly important. Desk-based workers spend approximately two-thirds of their daily working hours in a sitting position [8]. In addition, desk-based work and prolonged computer use are increasing due to the digitalization of systems in work environments and the handling of documents from computers [9]. Although there has been an increase in working from home styles, especially after the pandemic, the time spent at the desk is still similar to the time spent in the workplace [9 , 10]. Static muscle activity and improper postures maintained during prolonged computer use can lead to increased strain on cervical, upper extremity, and trunk muscles, increasing musculoskeletal symptoms [11]. This situation may lead to pain and functional limitations, especially in the head, neck, shoulders, wrists, and back regions [9 , 10 , 12 – 14]. Such musculoskeletal complaints are frequently reported among in desk-based workers [15 , 16], potentially contributing to increased disability and decreased work productivity. Frequent headaches are closely associated with common NP in desk-based workers [17]. For example, a study of desk-based workers reported that approximately 80% of individuals with headaches also experienced NP [16]. Long sitting times, poor sitting posture, static sitting, and less frequent breaks are associated with LBP in desk-based workers [8]. In addition, it has been shown that high workload-stress, time pressure, and psychosocial stress factors may be associated with both headaches and musculoskeletal pain in desk-based workers [18 – 21]. Therefore, desk-based work conditions may be an important environmental factor that can play a role in the onset or exacerbation of migraine and musculoskeletal symptoms [12 , 21]. The long-standing burden of headache disorders in neurology and global health has been addressed in recent years within the framework of Sustainable Development Goal 3 (SDG 3); it is considered a significant priority, particularly in terms of promoting healthy living, maintaining workforce participation, and improving quality of life [22]. Despite this, studies examining CS, work-related stress, and musculoskeletal symptoms variables together in desk-based workers with and without migraine are limited in the literature. The majority of existing studies have focused on individuals with headache and NP, and musculoskeletal regions have mostly been overlooked [6 , 7]. However, desk-based work conditions are associated with multi-regional loading affecting not only the cervical region but also the upper extremities and trunk. Accordingly, comparing desk-based workers with and without migraine could fill an important gap in the literature. The aim of this study were (1) to compare CS, work-related stress levels, and musculoskeletal pain symptoms in desk-based workers according to the presence of migraine; (2) to examine the associations among these variables; and (3) to identify factors independently associated with CS in individuals with migraine. In our first hypothesis, we assumed that desk-based workers with migraine would exhibit higher levels of CS, work-related stress, and multi-region musculoskeletal pain compared to desk-based workers without migraine. In our second hypothesis, we assumed that CS would show a correlation with GWSS, migraine burden, and musculoskeletal pain and that these variables would be independently associated with CS in individuals with migraine.

Methods

Study design and participants This study cross-sectional, observational study compared desk-based workers with and without migraine diagnosis. Ethical approval for this study was obtained from the Bingöl University Health Sciences Scientific Research and Publication Ethics Committee (Approval No: 3/10, and date: 02.04.2026). The study was registered at ClinicalTrials.gov with the number NCT07554664 (registration date: 21.04.2026) and was reported in accordance with the STROBE checklist (Supplement 1). The BioRender program was used for the Figure (<https://www.biorender.com/>). A total of 228 individuals were included in the study: 84 desk-based workers without a diagnosed migraine, 74 desk-based workers with EM, and 70 desk-based workers with CM. The face-to-face data collection and analysis process was conducted between April, 2026 and May 2026. Patients diagnosed with EM and CM by a neurologist at Gaziantep City Hospital, Department of Neurology, according to the International Classification of Headache Disorders Edition 3 (ICHD 3) [23], were included in the study. The analysis, interpretation, and article writing stages were carried out at Yozgat Bozok University, Faculty of Health Sciences, Department of Physiotherapy and Rehabilitation; Bingöl University, Faculty of Physical Therapy and

Rehabilitation; and Ankara Atatürk Sanatorium Training and Research Hospital, Department of Neurology. To minimize potential bias, eligible participants were invited to complete the questionnaires based on inclusion and exclusion criteria. Standardized instructions were provided, anonymity was ensured to encourage honest responses, and validated measurement tools were used. Inclusion and exclusion criteria of the study Inclusion criteria for the study were: being between 18 and 65 years of age, having been employed in an office job for at least 6 months, working in a desk-based occupation, being able to read and understand Turkish, agreeing to participate voluntarily, and for individuals with a migraine diagnosis, having an EM or CM diagnosis according to the ICHD 3 criteria [23]; for individuals without a migraine diagnosis, not having a diagnosis or history of migraine or other headache disorders and not reporting recurrent headache episodes during the previous year. Exclusion criteria for the study were: a history of neurological disorder (e.g., Multiple Sclerosis), inflammatory rheumatic disorders (e.g., Rheumatoid Arthritis), generalized chronic pain syndromes (e.g., Fibromyalgia), any other headache disease, any other systemic disease, acute fracture and infection, spinal surgery or spinal disc herniation within the last 6 months, serious musculoskeletal trauma within the last 6 months, pregnancy, a cognitive or psychiatric condition that would prevent completion of the questionnaires, and, in the CG without a migraine diagnosis, a history of recurrent headaches within the last year. Outcome measures Primary outcome measure Central sensitization The Central Sensitization Inventory (CSI) measures somatic and emotional symptoms common in CS syndrome. Section A measures 25 symptoms with 5 response options (0 to 4). The total score ranges from 0 to 100. Section B asks individuals whether they have previously been diagnosed with preliminary specific CSI conditions [24]. Secondary outcome measures Demographic and clinic evaluations Demographic and clinical information such as sex, age, weight, height, education level, occupation, medical history, family history, number of years diagnosed with migraine (if any), migraine triggers, symptoms experienced during attacks, number of headache days per month, number of headache attacks per month, duration of attacks, localization of headaches, and sitting time at a desk were recorded for all individuals. Headache intensity The Numerical Pain Rating Scale (NPRS) consists of pain levels where the person indicates the pain they are currently experiencing with a vertical line on a plane. The pain felt by the person is given on a scale of 0 to 10 as 0: no pain, 10: I feel very severe pain [25]. Disability The Migraine Disability Assessment (MIDAS) questionnaire evaluates the last 3 months of disability and consists of a 5-item self-administered test covering disability at work/school, household chores, family and social, or leisure activities. The total number of days missed in these activities is the total score and classifies the disability as minimal disability (0-5 points), mild disability (6-10 points), moderate disability (11-20 points), or severe disability (21+) [26]. Headache impact The Headache Impact Test-6 (HIT-6) is a questionnaire for headaches that assesses vitality, pain, psychological distress, sociability, role, and cognitive functioning. Each item is scored on a 5-point Likert scale (6 = never, 8 = rarely, 10 = sometimes, 11 = very often, 13 = always). The total score is determined by summing the scores on all six items and ranges from 36 to 78 points. (49 = little/no impact, 50-55 = some impact, 56-59 = significant impact, and 60-78 = serious impact) [27]. Work-related stress The General Work Stress Scale (GWSS) is a 9-item questionnaire that measures how much stress individuals experience in the workplace. It consists of 5 Likert-type items (Never = 0, Rarely = 1, Sometimes = 2, Mostly = 3, Always = 4). Higher scores indicate increased workplace stress [28]. Musculoskeletal pain symptoms The Nordic Musculoskeletal Questionnaire (NMQ) is a standardized, subjective data collection tool used to identify musculoskeletal disorders and pain in different areas of the body (low back, neck, shoulders, etc.). Individuals are asked whether they have experienced pain/discomfort in the relevant areas in the last 1 year or the last 7 days, usually coded as Yes/No. The total number of painful areas is determined [29]. Sample size calculation The primary outcome measure of the study was the CSI. However, since there were no previous studies evaluating the CSI in desk-based workers with and without migraine diagnosis, and comparable pilot data were not available, a priori sample size was calculated based on a moderate effect size [30]. A one-way analysis of variance (ANOVA) model was selected for comparing three independent groups (healthy control, EM, and CM) using the G*Power program (version 3.1) [30]. With an effect size of $f = 0.25$, an alpha error probability of 0.05, and a power of 80%, the required total sample size was calculated as 159 individuals [30]. The actual power value obtained for this sample size was 0.805. In total, 228 individuals were included in the study. The final sample size exceeded the required minimum sample size, thus enhancing the robustness of the study findings. Data analysis Statistical analyses were performed using IBM SPSS Statistics for Mac version 31.0 (IBM Corp., Armonk, NY, USA). Visual methods (histograms and probability plots) and the Kolmogorov-Smirnov test were used to assess the normality of demographic, clinical, and questionnaire variables. Normally distributed numerical variables were presented as mean $\pm$ standard deviation, non-normally distributed numerical variables as median (interquartile range), and categorical variables as number (n) and percentage (%). For comparisons between the three groups, ANOVA was used for normally distributed variables, and the Kruskal-Wallis test was used for non-normally distributed variables. In ANOVA and Kruskal-Wallis analyses, effect size was estimated using eta squared (η^2) and epsilon squared (ϵ^2), respectively, and interpreted as small 0.01, medium 0.06, and large 0.14 [31].

]. Post-hoc pairwise comparisons were performed using Bonferroni or GamesHowell tests, as appropriate. To examine whether group differences were significant after controlling for potential confounding factors, univariate general linear models were performed by including variables such as sex, occupational category, age, body mass index (BMI), daily sitting time, and years of experience in the current job as covariates. The strength of association for categorical comparisons was assessed using Cramers V (0.05: weak, 0.10: moderate, 0.15: strong, 0.25: very strong) [32]. For correlation analyses, the Pearson correlation coefficient was applied to normally distributed data, and the Spearman rank correlation coefficient was applied to non-normally distributed data (0.100.39: weak, 0.400.69: moderate, and 0.70: strong) [33]. Multiple linear regression analysis was performed to control for potential confounding factors and to determine factors associated with CS in the migraine group. The variables included in the regression model were selected based on clinical relevance and significant associations observed in univariate analyses. The GWSS and NMQ variables were included because they represented the primary constructs of the study. MIDAS and monthly headache days were included to reflect migraine-related disability and headache burden, while age and BMI were included as potential confounding variables commonly considered in pain-related studies. To avoid conceptual overlap between migraine burden measurements, HIT-6 was not included in the final regression model. Unstandardized regression coefficients (B), standardized coefficients (β), 95% confidence intervals (95% CI), t-values, Variance Inflation Factor (VIF) values, and p-values were reported. Regression assumptions, including residual normality, linearity, and multicollinearity, were evaluated using residual plots, normal probability plots, and VIF values. VIF values ≤ 5 were considered indicative of potential multicollinearity [34 , 35]. For all analyses, a p-value ≤ 0.05 was considered statistically significant.

Results

Demographic and clinical characteristics of participants

A total of 228 individuals (CG: n = 84, EM: n = 74, CM: n = 70) were included in the study. There were no significant differences between the groups in terms of age, BMI, daily sitting time, or years of work in the current job (p ≥ 0.05 for all values). Height differed significantly among groups (p = 0.026). Sex distribution differed significantly among the groups ($\chi^2(2) = 15.36$, p 0.001), and the proportion of females was significantly higher in the migraine groups than in the CG (Table 1). Furthermore, occupation category also differed significantly between the groups ($\chi^2(12) = 27.964$, p = 0.006, Cramers V = 0.248), indicating a small to moderate association between migraine status and occupational type (Supplementary Table S1).

Table 1 Demographic and clinical characteristics of participants

Variables	Control Group (n = 84)	Median (IQR) or n (%)	Episodic Migraine (n = 74)	Median (IQR) or n (%)	Chronic Migraine (n = 70)	Median (IQR) or n (%)	H or χ^2	p Value
Age (years)	34 (30-40)	34 (28-42)	38 (30-44)	2.960	0.228			
Sex	Female	Male	49 (58.3%)	35 (41.7%)	62 (83.8%)	12 (16.2%)	56 (80%)	14 (20%)
Weight (kg)	69.5 (60.25-82.75)	65 (58-73)	68.5 (59.75-78)	5.999	0.050			
Height (cm)	168 (160.25-175)	165 (160-170)	164 (159.75-170)	7.319	0.026*			
BMI (kg/m ²)	24.80 (22.42-27.05)	23.96 (21.75-26.13)	25.52 (22.18-28.56)	4.116	0.128			
Daily sitting time at the desk (hours)	8 (6-8)	8 (6-8)	6.5 (4-8)	2.806	0.246			
Years of work in the current job	8 (3-12)	9.5 (3-13)	7.55 (2-17)	0.331	0.848			

kg: kilogram; cm: centimeter; BMI: Body Mass Index; IQR: Interquartile Range; H: Kruskal-Wallis test statistic; p Value: Statistical significance set at p 0.05; * p 0.05; *** p 0.001

Group comparisons: central sensitization, work-related stress, and musculoskeletal pain symptoms

There were significant differences between groups in terms of CSI (F (2,225) = 38.63, p 0.001, $\eta^2=0.256$). Post-hoc analyses showed that both EM and CM groups had significantly higher CSI scores than the CG (p 0.001 for both) (Table 2). This difference remained significant even after adjusting for sex, occupation category, age, BMI, daily sitting time, and years of work (F (2,214) = 30.446, p 0.001, partial $\eta^2=0.222$). Sex was also independently associated with CSI scores (F (1,214) = 8.578, p = 0.004). Work-related stress scores also differed significantly among groups (F (2,225) = 10.15, p 0.001, $\eta^2=0.083$). Both EM and CM groups had higher GWSS scores than the CG (p 0.001), but no significant difference was observed between EM and CM groups (p 0.05) (Table 2). This group effect remained significant even after adjustment for the same potential confounding factors above (F (2,214) = 14.342, p 0.001, partial $\eta^2=0.118$).

Table 2 Group comparisons of central sensitization and work-related stress scores

Variables	Control Group (n = 84)	Mean $\pm$ SD	Episodic Migraine (n = 74)	Mean $\pm$ SD	Chronic Migraine (n = 70)	Mean $\pm$ SD	F (2,225)	Overall p	η^2	Post-hoc comparisons
CSI	32.07 $\pm$ 14.77	46.58 $\pm$ 13.72	52.56 $\pm$ 16.08	38.63	0.001***	0.256				EM CG (p 0.001), CM CG (p 0.001)
GWSS	11.08 $\pm$ 7.35	15.74 $\pm$ 7.58	16.57 $\pm$ 9.52	10.15	0.001***	0.083				EM CG (p 0.001), CM CG (p 0.001)

CSI: Central Sensitization Inventory; GWSS: General Work Stress Scale; EM: Episodic Migraine; CM: Chronic Migraine; CG: Control Group; SD: Standard Deviation; F: ANOVA test statistic; p Value: Statistical significance set at p 0.05; *** p 0.001; Bonferroni post-hoc test was used for CSI, while GamesHowell post-hoc test was applied for GWSS due to violation of homogeneity of variance

Significant overall group differences were found for the number of painful body regions during the past 12 months (H (2) = 26.938, p 0.001), interfering with work during the past 12 months (H (2) = 10.515, p = 0.005), and during the past 7 days (H (2) = 39.180, p 0.001). Post-hoc analyses showed that the CM group reported significantly more painful body regions than CG for the number of painful body regions during the past 12 months (p = 0.004) and painful body regions interfering with work during the past 12 months (p = 0.001). For the number of painful body regions during the past 7

days, both the EM and CM groups reported significantly more painful body regions than CG ($p = 0.001$ for both). However, no significant difference was observed between EM and CM groups ($p = 0.05$) (Table 3). After adjustment for age, sex, BMI, occupation category, daily sitting time, and years of work group differences remained significant for the number of painful body regions during the last 12 months: $F(2,214) = 10.591$, $p = 0.001$, partial $\eta^2 = 0.090$, interfering with work during the past 12 months: $F(2,214) = 5.783$, $p = 0.004$, partial $\eta^2 = 0.051$, and during the past 7 days ($F(2,214) = 14.599$, $p = 0.001$, partial $\eta^2 = 0.120$). The most frequently reported musculoskeletal pain complaints in all groups were NP and LBP. The prevalence of NP over the 12 months was significantly higher in the migraine groups (EM: 83.8% and CM: 85.7%) compared to the CG (60.7%). Detailed distributions of painful body regions were presented in the Fig. 1 .

Table 3 Group comparisons of musculoskeletal pain regions

Variables	Control Group ($n = 84$)	Median (IQR)	Episodic Migraine ($n = 74$)	Median (IQR)	Chronic Migraine ($n = 70$)	Median (IQR)	H (2)	p	η^2
Number of painful body regions during the past 12 months	3 (1.25-4)	4 (36)	4 (2-6.25)	26.938	0.001***	0.111	CM CG ($p = 0.004$)		
Number of painful body regions interfering with work during the past 12 months	1 (12)	2 (13)	2 (14)	10.515	0.005**	0.038	CM CG ($p = 0.001$)		
Number of painful body regions during the past 7 days	1 (12)	3 (24)	3 (24)	39.180	0.001***	0.165	CM CG ($p = 0.001$)		

EM CG ($p = 0.001$)

NMQ: Nordic Musculoskeletal Questionnaire; EM: Episodic Migraine; CM: Chronic Migraine; CG: Control Group; IQR: Interquartile Range; H: Kruskal-Wallis test statistic, η^2 : Epsilon squared effect size (0.01 = small, 0.06 = medium, 0.14 = large); p Value : Statistical significance set at $p = 0.05$; ** $p = 0.01$; *** $p = 0.001$; Bonferroni post-hoc test was used for NMQ

Fig. 1 Distribution of musculoskeletal pain in controls (a), episodic migraine (b), and chronic migraine (c) groups according to the nordic musculoskeletal questionnaire. NMQ: Nordic Musculoskeletal Questionnaire

Comparison between episodic and chronic migraine

When comparing migraine groups, individuals with CM had significantly higher numbers of headache days ($p = 0.001$), attack duration ($p = 0.016$), pain intensity ($p = 0.009$), headache-related disability (MIDAS, $p = 0.001$), and headache impact (HIT-6, $p = 0.001$) compared to the EM group. There was no significant difference between the groups in terms of year of migraine diagnosis ($p = 0.055$) (Table 4).

Table 4 Comparison of disease symptoms between episodic and chronic migraine

Variables	Episodic Migraine ($n = 74$)	Median (IQR)	Chronic Migraine ($n = 70$)	Median (IQR)	U	Z	p	r
Migraine diagnosis years	5 (1-13.5)	8 (2.75-15)	2081.5	-1.921	0.055	0.16		
Number of headache days in the last month	5(37)	15 (15-17.25)	23.50	-10.331	0.001***	0.86		
Duration of headache attacks (days)	1 (0.2-3)	1.5 (0.87-3)	1993.0	-2.410	0.016*	0.20		
Headache intensity (NPRS)	7 (68)	8 (79)	1945.0	-2.616	0.009**	0.22		
Headache disability (MIDAS)	18 (1232)	32.5 (16-58.5)	1689.5	-3.600	0.001***	0.30		
Headache impact (HIT-6)	63 (6066)	66 (6272)	1541.0	-4.201	0.001***	0.35		

NPRS: Numeric Pain Rating Scale; MIDAS: Migraine Disability Assessment Scale; HIT-6: Headache Impact Test-6; IQR: Interquartile Range; The Mann-Whitney U test was used. U = MannWhitney U test statistic; Z = standardized test statistic; r = effect size; p : Statistical significance set at $p = 0.05$; * $p = 0.05$; ** $p = 0.01$; *** $p = 0.001$

Correlation analyses

In all groups, CSI scores showed positive correlations with GWSS and NMQ-3items ($p = 0.05$) (Table 5). In the EM and CM groups, CSI showed significant positive correlations with GWSS, MIDAS, HIT-6, and the number of painful body regions ($p = 0.05$) (Table 5 , Supplement Table S2).

Table 5 Correlation of variables in the control group, episodic and chronic migraine groups

Variable 1	Variable 2	ρ	p
Control Group CSI	GWSS	0.593	0.001***
CSI NMQ / Number of painful body regions during the past 12 months		0.458	0.001***
CSI NMQ / Number of painful body regions interfering with work during the past 12 months		0.332	0.002**
CSI NMQ / Number of painful body regions during the past 7 days		0.439	0.001***
Episodic Migraine CSI	Number of headache days in the last month	0.263	0.023*
CSI MIDAS		0.404	0.001***
CSI HIT-6		0.245	0.035*
CSI GWSS		0.466	0.001***
CSI NMQ / Number of painful body regions during the past 12 months		0.403	0.001***
CSI NMQ / Number of painful body regions during the past 7 days		0.287	0.013*
Chronic Migraine CSI	Duration of headache attacks (days)	0.353	0.003**
CSI NPRS		0.346	0.003**
CSI MIDAS		0.482	0.001***
CSI HIT-6		0.446	0.001***
CSI GWSS		0.518	0.001***
CSI NMQ / Number of painful body regions during the past 12 months		0.388	0.001***
CSI NMQ / Number of painful body regions interfering with work during the past 12 months		0.332	0.005**
CSI NMQ / Number of painful body regions during the past 7 days		0.369	0.002**

NPRS: Numeric Pain Rating Scale; MIDAS: Migraine Disability Assessment Scale; HIT-6: Headache Impact Test-6; CSI: Central Sensitization Inventory; GWSS: General Work Stress Scale; NMQ: Nordic Musculoskeletal Questionnaire; ρ : Spearman's rank correlation coefficient; p : Statistical significance set at $p = 0.05$; * $p = 0.05$; ** $p = 0.01$; *** $p = 0.001$

Variables independently associated with central sensitization

Multiple linear regression modeling focused on CSI because it is a key element associated with the relationship between migraine, work-related stress, and musculoskeletal pain symptoms. Multiple linear regression analysis was performed to identify factors associated with CSI in the migraine group. The overall model was significant ($F(6,137) = 20.514$, $p = 0.001$) and explained 47.3% of the variance in CSI scores ($R^2 = 0.473$, Adjusted $R^2 = 0.450$) (Table 6). Work-related stress ($\beta = 0.380$, $p = 0.001$), number of painful body regions in the last 1 year ($\beta = 0.382$, $p = 0.001$), headache-related disability (MIDAS) ($\beta = 0.230$, $p = 0.002$), and number of headache days per month ($\beta = 0.170$, $p = 0.016$) were identified as variables independently associated with CSI scores. Age and BMI were not associated with CSI scores ($p = 0.05$) (Table 6).

Table 6 Factors associated with

central sensitization in the migraine group: multiple linear regression analysis Variables (n = 144) B 95% CI t p VIF
 GWSS 0.674 0.444 to 0.903 0.380 5.810 0.001*** 1.113 NMQ / Number of painful body regions during the past 12 months 2.178 1.469 to 2.887 0.382 6.077 0.001*** 1.025 MIDAS 0.106 0.040 to 0.172 0.230 3.166 0.002** 1.367
 Monthly Headache days 0.423 0.080 to 0.767 0.170 2.436 0.016* 1.272 Age (years) -0.034 -0.238 to 0.171 -0.021 -0.326 0.745 1.116 BMI (kg/m²) 0.072 -0.403 to 0.548 0.020 0.300 0.764 1.116
 GWSS: General Work Stress Scale; NMQ: Nordic Musculoskeletal Questionnaire; MIDAS: Migraine Disability Assessment Scale; BMI: Body Mass Index; B = unstandardized regression coefficient; = standardized regression coefficient; t = t statistic; VIF : Variance Inflation Factor; p : Statistical significance set at p 0.05; * p 0.05; ** p 0.01; *** p 0.001

Discussion This study demonstrated that desk-based workers with migraine had higher CSI scores, greater GWSS, and more widespread musculoskeletal pain symptoms compared to desk-based workers without migraine. CS was more pronounced in CM, while GWSS and musculoskeletal pain symptoms did not differ between EM and CM. CS, GWSS, and widespread musculoskeletal pain symptoms were associated with each other. Additionally, GWSS, number of painful body regions during the past 12 months, disability, and number of headache days per month were variables independently associated with CSI scores. These findings suggest that migraine in desk-based workers is associated with both neurophysiological susceptibility and increased physical and psychosocial burden.

Demographic and clinical characteristics of participants As the three groups were similar in age, BMI, desk sitting time, and years in the current job, the observed significant difference in terms of female sex may be related to the clinical condition of migraine rather than the underlying occupational exposure. The similarity in desk sitting time is particularly important because prolonged sitting, paperwork, or computer use has been identified as an independent risk factor for musculoskeletal symptoms [36]. Therefore, it is understood that the differences in musculoskeletal symptoms in our study cannot be explained solely by sitting time. The proportion of office and self-employed workers in the CM group was noteworthy. This finding suggests that occupational factors and work environment may be important in migraine management. However, due to the cross-sectional design of the study, a causal interpretation cannot be made. Indeed, it has been reported that cervical and upper extremity pain is common in sitting and computer-based jobs, but individual and psychosocial factors also make significant contributions [11 , 37 39]. Accordingly, the fact that the groups had similar desk sitting times suggests that migraine constitutes an additional burden. The difference in sex distribution between the groups and the higher proportion of women in the migraine groups is an expected result. It has been known for many years that migraine is more common in women than in men. Global Burden of Disease data also show that migraine is a significant cause of disability, especially in women of working age. Therefore, the higher proportion of women in the migraine groups in our sample is consistent with clinical epidemiology [40].

Comparison of central sensitization, work-related stress, and musculoskeletal pain symptoms among groups One of the key findings of our study is that CSI scores were higher in both the EM and CM groups compared to the CG. Furthermore, the higher CSI scores in the CM group compared to the EM group suggest that CS may be associated with a higher disease burden and symptom severity. This indicates that individuals with CM may need multidisciplinary approaches targeting not only attack frequency but also pain modulation systems. Similarly, the more widespread musculoskeletal pain symptoms observed in the migraine groups also support the clinical significance of CS. Previous studies have also reported that CSI scores are highest in CM, followed by EM, and lowest in healthy individuals [41 , 42]. CS is characterized by increased excitability in the central nervous system and is associated with cutaneous allodynia, widespread pain, increased attack severity, delayed treatment response, and psychosocial burden [2 , 3].

General Work Stress Scale scores were higher in both migraine groups compared to the CG; this is consistent with previous findings that psychosocial workload, time pressure, and decreased job control may be associated with greater migraine burden and impair productivity [43 45]. Furthermore, the lack of migraine-friendly work environments among healthcare workers has been reported to increase the migraine burden [1]. Therefore, it is emphasized that suitable work environments for migraine can be created by considering facilitating and inhibiting factors [43]. However, as work-related stress did not differ between EM and CM, stress may contribute to the overall migraine burden but does not explain chronicity alone. Beyond statistical significance, many of the observed differences also appear to be clinically significant. CSI scores in the migraine groups were significantly higher than those in the CG and approached levels associated with clinically significant CS symptoms. Similarly, MIDAS and HIT-6 scores in the CM group reflect a significant disability associated with headache and its impact on daily functioning. In addition, the greater number of painful body regions reported in the migraine groups demonstrates a clinically significant burden of widespread musculoskeletal pain symptoms. These findings support the potential clinical importance of routinely assessing CS, work-related stress, and musculoskeletal pain symptoms in desk-based workers with migraine. The higher number of painful body regions observed in migraine groups suggests that migraine extends beyond headache and may be associated with multi-region musculoskeletal symptoms. The relationship between migraine and musculoskeletal pain can be explained by several mechanisms. First, due to CS, the pain threshold is reduced, and

somatic inputs from different body regions are perceived more intensely [46 , 47]. Second, peripheral sensitization, protective postures, changes in muscle activity, and avoidance behaviors that develop during migraine attacks can lead to secondary musculoskeletal problems [46]. Third, desk-based work conditions may be associated with greater musculoskeletal symptoms in individuals with migraine by creating prolonged static load [17]. Previous studies also support the fact that neck, lower back, and shoulder pain are more frequently observed in individuals with migraine, and lower extremity pain is reported less frequently [47 , 21 , 48-54]. NP was the most common complaint in all groups and was significantly more frequent in individuals with migraine. This finding is consistent with previous reports showing a significantly higher prevalence of NP in migraine, particularly in CM [4-6]. Functional and anatomical connections within the trigeminocervical complex may explain this association [5]. The higher incidence of LBP and shoulder pain in migraine groups also supports widespread pain processing disorders [7 , 48-52]. Lower extremity pain in migraine patients has mostly been reported at the case report level and, while noteworthy in terms of central pain processing mechanisms, requires further research [53]. Overall, the most frequently reported symptoms are seen in the neck, lower back, shoulder, and upper back regions [21 , 54]. However, pain is also reported with considerable frequency in the upper extremity (hand and wrist) and the lower extremity (knee, ankle, and thigh) regions [54]. Prolonged sitting during desk work, combined with fatigue in the trunk muscles, repetitive use of body parts for tasks, and postural loading, can be expected to increase pain symptoms in migraine patients who already have a predisposition to CS. As the number of painful musculoskeletal regions did not differ between the EM and CM groups, widespread body pain appears to be more closely associated with the presence of migraine, whereas chronicity may relate more to central pain processing than to the number of painful regions. Comparison of episodic and chronic migraine groups The significantly higher number of headache days per month, attack duration, headache intensity, MIDAS, and HIT-6 scores in the CM group compared to EM are expected results that support the diagnostic classification. CM is characterized by a more frequent number of headache days by definition. However, the increase not only in frequency but also in attack duration and pain intensity indicates that the disease burden is multidimensional [55]. High MIDAS scores indicate that individuals with CM experience more disability in their work, home, and social lives [56]. Similarly, the high HIT-6 scores in CM suggest that the impact of headache on quality of life is more pronounced [57]. The lack of difference in terms of migraine diagnosis duration suggests that the current clinical burden and attack frequency may be more decisive than how long the disease has been present. Previous studies have reported that high attack frequency, drug overuse, obesity, depression, sleep disorders, and other comorbid conditions are important risk factors for migraine progression [58]. However, disease duration alone does not stand out as a consistent predictor. In other words, receiving a migraine diagnosis for many years alone may not explain chronicity. Correlation analyses In our study, positive correlations were found between CS and GWSS and musculoskeletal pain symptoms (NMQ) in all groups. This finding shows that the pain experience is related not only to peripheral nociception but also to psychosocial and behavioral factors associated with work-related stress. In particular, the relationships between CSI and MIDAS, HIT-6, and the number of painful areas in migraine groups suggest that disease burden, functional loss, and impairment in quality of life may be more pronounced in individuals with high CSI scores. Similarly, it has been reported that patients with chronic headaches and, to a lesser extent, EM show lower pain thresholds, more widespread somatic sensitization, and higher CS symptoms compared to healthy controls [59]. Clinical studies on migraines have also reported that an increase in CS is associated with higher headache-related disability, and that improvement in CS after exercise and muscle relaxation treatments is seen together with improvements in MIDAS and HIT-6 scores [60-62]. It is noteworthy that the relationship between CSI and GWSS is stronger in the CM group. This may reflect the bidirectional interaction between stress systems and central pain modulation [63 , 64]. Long-term stress disrupts pain modulation mechanisms via the hypothalamo-pituitary-adrenal axis and the autonomic nervous system, while chronic pain can increase stress perception [64 , 65]. This reciprocal interaction may contribute to an increase in the severity and frequency of headaches accompanying persistent pain, the development of allodynia, and the maintenance of CS. Continuous sympathetic activity, on the other hand, can support a generalized pain frame by increasing muscle tone, especially in the head, neck, and shoulder region [65]. The correlation of musculoskeletal symptoms with each other also suggests a generalized pain burden. Individuals experiencing pain in one region are more likely to report pain in other regions as well. It has also been shown that neck, back, and shoulder pain cluster together, and multisite pain patterns are common in desk-bound populations [54]. This situation demonstrates that focusing on only a single area in clinical assessment may be insufficient for addressing pain management in desk-based workers with migraine. Factors associated with central sensitization One of the key findings of our study is that the factors independently associated with CS in the regression analysis were GWSS, number of painful body regions during the past 12 months, MIDAS score, and number of headache days per month. The model explains approximately half of the variance showing that CS has strong and multidimensional determinants. The number of painful areas is an independent determinant that supports the close

relationship between generalized somatic pain and CS [65]. Similarly, it has been reported that individuals with migraine with generalized body pain have higher CSI scores and worse functional status [60]. The contribution of the number of headache days per month to the model suggests that recurrent nociceptive load may lead to permanent changes in the central nervous system. Indeed, population-based cohort studies have emphasized that an increase in headache frequency is one of the strongest predictors of the transition from EM to CM [58]. The independent contribution of the MIDAS score shows that functional disability may not only be a result but also a factor maintaining CS [56]. The independent effect of work stress reveals the importance of the psychosocial environment on neurophysiological pain processes. In a study conducted on an academic staff population, it was found that 92.6% of the participants experienced pain in at least one body region per year, 74% experienced pain weekly, 64.4% experienced pain in three or more regions, and musculoskeletal pain was associated with decreased work performance [66]. These findings are important in clinical practice. In addition to focusing on pharmacological attack control in migraine treatment, multi-faceted approaches based on SDG 3, such as managing workplace stress that may be associated with greater migraine burden, widespread adoption of migraine-friendly workplaces, ergonomic adjustments, and avoiding prolonged sitting, exercise therapy, sleep regulation, and pain education, can also be effective [22].

Clinical implications and future directions Expanding on our findings with clinical implications, recent pharmacological and non-pharmacological evidence suggests that there may be benefits for CS symptoms. Among pharmacological interventions, calcitonin gene-related peptide monoclonal antibodies may improve in the CSI, Allodynia Symptom Checklist-12 and migraine-related disability, pain intensity, duration, pain catastrophizing, anxiety-depression scores after treatment, suggesting a potential improvement in CS-related symptom burden in migraine [67–69]. Exercise and physical therapy interventions have been reported to reduce migraine symptoms, pain intensity, medication use, and disability, but the certainty of the evidence varies, and the optimal type of intervention remains unclear. Furthermore, the use of aerobic exercise and progressive muscle relaxation techniques with prophylactic medication has shown improvements in migraine-related outcomes in the clinical studies [70–74]. Therefore, high CSI, GWSS, and NMQ scores may help identify desk-based workers with migraine and could benefit from a multidisciplinary management approach that includes optimizing migraine preventive treatments, assessing medication overuse and allodynia, implementing structured physical therapy and exercise programs, ergonomic interventions, stress management, sleep assessment, and workplace adjustments. The findings of our study are significant from the perspective of SDG 3 (ensure healthy lives and promote well-being for all at all ages), which emphasizes reducing disability burden and improving quality of life [22]. We recommend that the assessment process for headache in desk-based workers with migraine should not be limited to headache frequency and severity, but should also routinely include questioning the level of CS, work-related stress, and multi-regional musculoskeletal symptoms. Multidisciplinary rehabilitation programs may be beneficial, especially in individuals with CM, as CS is prominent [75]. Future studies may aim for prospective longitudinal study designs to better understand and interpret causal relationships. The use of objective measurement methods such as pressure pain threshold and quantitative sensory test may strengthen the findings. In addition, ergonomic risk factors such as screen exposure, break frequency, sitting posture, and work environment can be additionally examined in future studies. Intervention studies, stress management programs in workplaces, and studies on migraine-friendly workplaces may be given importance. Additionally, future longitudinal studies using more advanced analytical approaches, such as Structural Equation Modeling [76, 77], may help clarify the complex direct and indirect relationships among CSI, GWSS, and NMQ. Finally, studies on different occupational groups, age ranges, and cultures/countries can also increase generalizability. Including non-desk workers in future studies could help clarify whether the observed findings are primarily related to occupational exposure or to migraine-specific mechanisms.

Limitations This study has several limitations. First, the use of self-report questionnaires and the fact that most individuals with migraine were recruited from a hospital setting may have led to recall and selection bias. Second, objective measures related to CS or pain processing, such as quantitative sensory testing, pressure pain thresholds, temporal summation, and conditioned pain modulation, ergonomics, activity level, and psychosocial factors were not performed. Therefore, CSI findings should be interpreted cautiously, as CSI is a screening tool and does not directly confirm CS as a neurophysiological mechanism. Third, relevant confounding factors such as psychosocial factors, medication use, physical activity level, cutaneous allodynia (such as Allodynia Symptom Checklist), and hormonal factors were not adequately assessed. These factors may have influenced CS results. Additionally, although relevant variables such as sex and occupational subgroup were statistically adjusted, residual confounding factors may still be present. Fourth, the cross-sectional design of the study precludes causal inference regarding the relationships between migraine, CS, GWSS, and NMQ. Although regression analyses allowed the investigation of independent associations, the directionality of these relationships remains unclear. Finally, because the sample consisted only of desk-based workers, the generalizability of the findings to other occupational groups may be limited.

Conclusion Central sensitization, work-related stress, and musculoskeletal pain

symptoms are higher in desk-based workers with migraine compared to desk-based workers without migraine. The key variable differentiating chronic migraine from episodic migraine is the more pronounced central sensitization. These findings highlight the need for multidimensional management approaches that address headache burden, psychosocial stress, and generalized pain in desk-based workers. Supplementary Information Below is the link to the electronic supplementary material. Supplementary Material 1